

Concept Question 9.1 PDS 70 protoplanetary disk

Fig. 1 shows six distinct simulations of the protoplanetary disk around PDS 70 and the corresponding SED.

- In the lecture we discussed different disk classifications. What kind of disk is depicted in Fig. 1?
- Which legend is the correct one (Legend 1 or Legend 2)? Explain your answer. *Hint:* Only one parameter is varied in the different simulations. Which one?

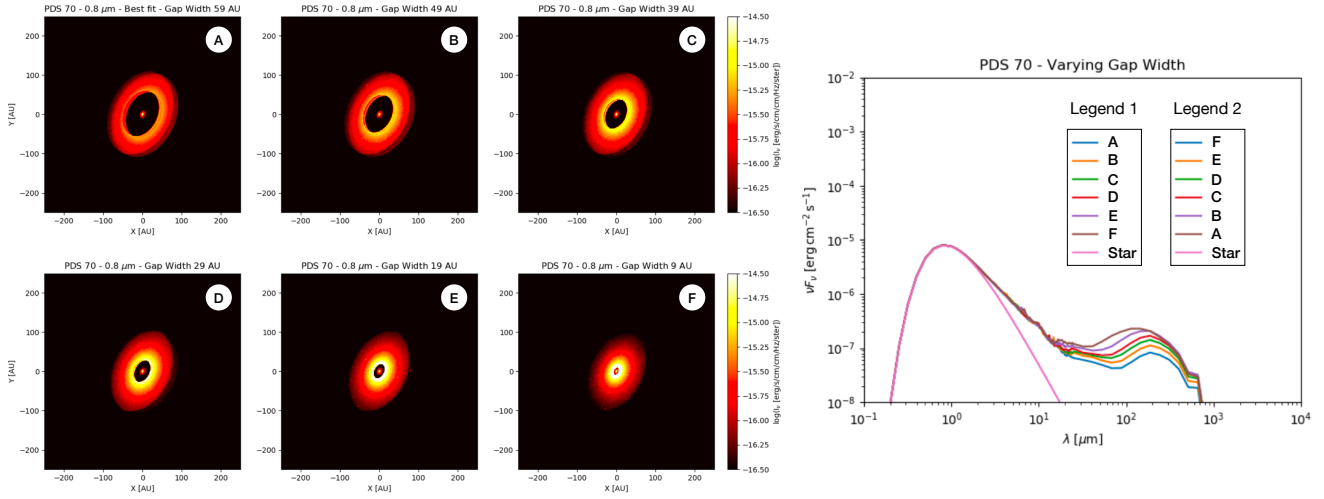


Figure 1: Simulated disk structure and SED for PDS 70. Credit: V. Faramaz (adapted).

Problem 9.2 Toomre Stability in Protoplanetary Disks

- Starting from the dispersion relation for axisymmetric perturbations in a thin, self-gravitating gaseous disk

$$\omega^2 = \kappa^2 + k^2 c_s^2 - 2\pi G \Sigma |k|, \quad (1)$$

derive the local Toomre stability condition for perturbations at all wavelengths, and show that it can be written as

$$Q \equiv \frac{c_s \kappa}{\pi G \Sigma} \geq 1. \quad (2)$$

- For a protoplanetary disk at $r = 30$ AU with $\Sigma = 100$ g cm⁻² and $T = 50$ K, compute c_s and Ω , then evaluate Q (assume a Keplerian disk around a $1 M_\odot$ star so that $\kappa \simeq \Omega$).

- c) Physically interpret the unstable wavelength range in terms of potential fragmentation or spiral modes in a protoplanetary disk. What is the effect of rotation, pressure, and self-gravity?

Problem 9.3 Hydrostatic Equilibrium

In the vertical direction, the disc structure can be described by assuming hydrostatic equilibrium. Further assuming that the star's gravity dominates over the entire disk, we find:

$$\frac{1}{\rho} \frac{\partial P}{\partial z} + \left(\frac{GM_{\star}}{a^2} \right) \left(\frac{z}{a} \right) = 0$$

where G is the gravitational constant, $M_{\star}$ is the mass of the star, and a is the orbital distance.

- a) What expression can be found for P in dependency of ρ assuming an ideal gas? Use it to substitute for the term dependent on P in the above equation.
- b) Use a reasonable expression for ρ in dependency of the disk scale height H to find an expression for the scale height H .

Problem 9.4 49 Ceti debris disk

In this exercise we will analyse spectroscopic and photometric data of 49 Ceti, an A1V star at $d = 59$ pc. This star has an intriguing disk, exhibiting both the dust properties of a debris disk and the gas properties of a low-mass protoplanetary disk.

- a) Read in and plot the total SED of the 49 Ceti system (`SED_49_Cet.dat`). What part of the electromagnetic spectrum (e.g. X-ray, optical, UV) do the observations belong to? What features can you identify?
- b) Previous studies have shown that 49 Ceti is an A1V star with $T = 9000$ K. The corresponding ATLAS9 stellar model (Castelli & Kurucz 2004) is provided with this exercise sheet (`ATLAS9_9000_40.dat`). Read in the model, normalise it to fit the stellar part of the SED, and use it to estimate the stellar radius.
- c) Fit the disk part of the SED using a modified black body model:

$$F_{\nu}(T) = B_{\nu}(T)(1 - e^{-\tau_{\nu}})\Omega \approx B_{\nu}(T)\tau_{\nu}\Omega \quad (3)$$

where the solid angle Ω can simply be treated as a normalisation constant and the optical depth τ_{ν} is parametrised as:

$$\tau_{\nu} = \begin{cases} \tau_0 & \text{if } \lambda < \lambda_0 \\ \tau_0 \left(\frac{\lambda}{\lambda_0} \right)^{-\beta} & \text{if } \lambda > \lambda_0 \end{cases} \quad (4)$$

Which temperature do you find? *Hint:* Use Wien's law to estimate the initial temperature for the fit.

- d) Sum the models you've obtained for the stellar and disk parts of the SED to get the overall model, and plot it against the data. Does the model fit the whole SED well? If not, what does this tell you? What is missing?
- e) Based on the models that you obtained above, compute the fractional infrared luminosity $L_{\text{dust}}/L_{\star}$. This is defined as the ratio of the black body model integrated over all frequencies to the similarly integrated stellar model.
- f) The file `CII.data.dat` contains the *Herschel* spectrum around the [CII] $158\,\mu\text{m}$ emission line. Compute the total CII mass for the cases $T_{\text{gas}} = (20, 100, 1000)$ K using the relation:

$$M_{\text{CII}} = \frac{4\pi\lambda_0}{hc} \frac{F_{\text{int}} m d^2}{A_{\text{ul}} x_u} \quad (5)$$

where λ_0 is the wavelength of the line, F_{int} its integrated flux, A_{ul} is the spontaneous transition probability and x_u the fraction of atoms in the upper energy level. Based on sub-mm CO emission, Hughes et al. (2008) found $M_{\text{CII}} = 2.2 \times 10^{-4} M_{\oplus}$. Is this consistent with your result?

Hint: Assume $A_{\text{ul}} x_u = (4.06 \times 10^{-8}, 9.93 \times 10^{-7}, 1.45 \times 10^{-6})$ for $T_{\text{gas}} = (20, 100, 1000)$ K, respectively.